

DSC 40B

Lecture 4-5 :

**Big- Theta, Big-Oh
and Omega,
Formalized**

News

- New look for DSC40B



Agenda

Plan for the lecture

- **Formally** define Θ , O , Ω notation.
- Some useful properties.

***Formally define Θ , O , Ω
notations***

So Far

- Time Complexity Analysis: a picture of how an algorithm **scales**.
- Can use Θ -notation to express time complexity.
- Allows us to **ignore** details in a rigorous way:
 - **Saves us work!**
 - **But what exactly can we ignore?**

Theta Notation, Informally

- $\Theta(\cdot)$ **forgets** constant factors, lower-order terms.

$$5n^3 + 3n^2 + 42 = \Theta(n^3)$$

Theta Notation, Informally

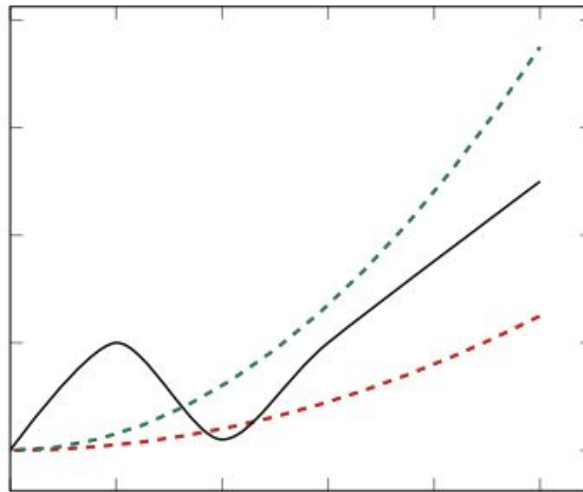
- $f(n) = \Theta(g(n))$ if $f(n)$ “grows like” $g(n)$.

$$5n^3 + 3n^2 + 42 = \Theta(n^3)$$

Definition

We write $f(n) = \Theta(g(n))$ if there are **positive** constants N , c_1 and c_2 such that for all $n \geq N$:

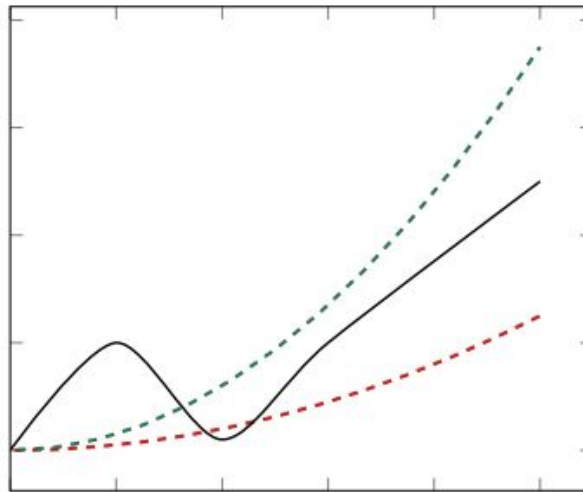
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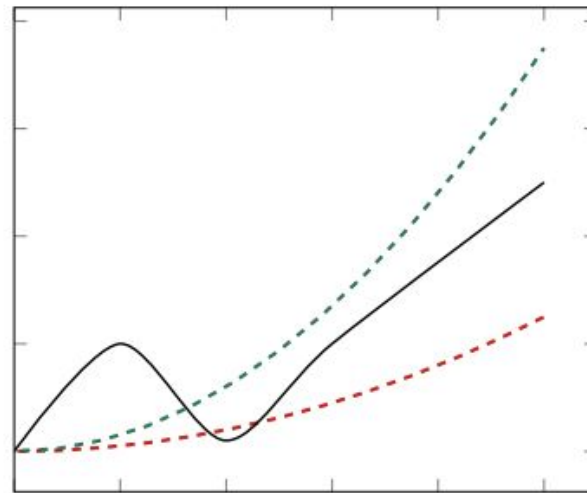
$$c_2 \cdot g(n)$$

$$c_1 \cdot g(n)$$

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$$c_2 \cdot g(n)$$

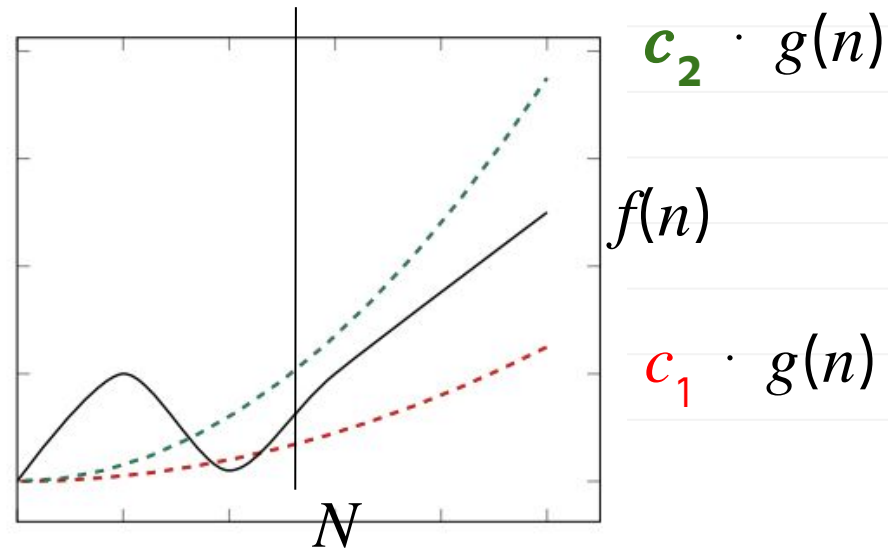
$$f(n)$$

$$c_1 \cdot g(n)$$

Definition

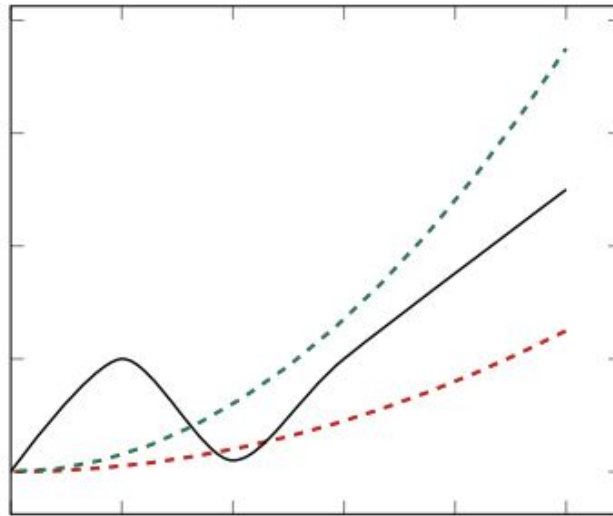
We write $f(n) = \Theta(g(n))$ if there are **positive** constants N , c_1 and c_2 such that for all $n \geq N$:

$$c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$$



Main Idea

If $f(n) = \Theta(g(n))$, then when **n is large** f is “*sandwiched*” between copies of g .



Proving Big-Theta

- We can prove that $f(n) = \Theta(g(n))$ by finding these constants.

$$c_1 g(n) \leq f(n) \leq c_2 g(n) \quad (n \geq N)$$

- Requires a **lower bound** and an **upper bound**.

Strategy: Chains of Inequalities

- To show $f(n) \leq c_2 g(n)$, we show:

$$f(n) \leq (\text{something}) \leq (\text{another thing}) \leq \dots \leq c_2 g(n)$$

- At each step:
 - We can do anything to make value **larger**.
 - But the goal is to simplify it to look like $g(n)$.

Example

- Show that $4n^3 - 5n^2 + 50 = \Theta(n^3)$
- Find constants c_1 , c_2 , N such that for all $n > N$:
$$c_1 n^3 \leq 4n^3 - 5n^2 + 50 \leq c_2 n^3$$
- They don't have to be the "best" constants! **Many solutions!**

Example

$$c_1 n^3 \leq 4n^3 - 5n^2 + 50 \leq c_2 n^3$$

- We want to make $4n^3 - 5n^2 + 50$ “look like” cn^3 .
- For the upper bound, can do *anything* that makes the function **larger**.
- For the lower bound, can do *anything* that makes the function **smaller**.

Example (n is a positive integer)

$$c_1 n^3 \leq 4n^3 - 5n^2 + 50 \leq c_2 n^3$$

- Upper bound:

$$4n^3 - 5n^2 + 50 \leq 4n^3 + 50 \leq 4n^3 + 50n^3 = 54n^3$$

Upper-Bounding Tips

- “Promote” lower-order **positive** terms:

$$3n^3 + 5n \leq 3n^3 + 5n^3$$

- “Drop” **negative** terms:

$$3n^3 - 5n \leq 3n^3$$

Example

$$c_1 n^3 \leq 4n^3 - 5n^2 + 50 \leq c_2 n^3$$

- Lower bound:

$$4n^3 - 5n^2 + 50 \geq 4n^3 - 5n^2$$

$$\geq 3n^3 + (n^3 - 5n^2) \geq 3n^3 \quad (n \geq 5)$$

True WHEN $(n^3 - 5n^2) \geq 0 \Rightarrow n \geq 5$

Lower-Bounding Tips

- “Drop” lower-order **positive** terms:

$$3n^3 + 5n \geq 3n^3$$

- “Promote and cancel” **negative** lower-order terms if possible:

$$4n^3 - 2n \geq 4n^3 - 2n^3 = 2n^3$$

Lower-Bounding Tips

- “Cancel” negative lower-order terms with big constants by “breaking off” a piece of high term.

$$4n^3 - 10n^2 = (3n^3 + n^3) - 10n^2$$

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$$n^3 - 10n^2 \geq 0 \text{ when}$$

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$$n^3 - 10n^2 \geq 0 \text{ when } n^3 \geq 10n^2 \Rightarrow n \geq 10:$$

Lower-Bounding Tips

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$$4n^3 - 10n^2 = (3n^3 + n^3) - 10n^2 = 3n^3 + (n^3 - 10n^2)$$

$$n^3 - 10n^2 \geq 0 \text{ when } n^3 \geq 10n^2 \Rightarrow n \geq 10:$$

$$\geq 3n^3 + 0 \text{ (} n \geq 10 \text{)}$$

Example

$$c_1 n^3 \leq 4n^3 - 5n^2 + 50 \leq c_2 n^3$$

- We want to make $4n^3 - 5n^2 + 50$ “look like” cn^3 .
- For the upper bound, can do anything that makes the function **larger**.
- For the lower bound, can do anything that makes the function **smaller**.

Example

$$3n^3 \leq 4n^3 - 5n^2 + 50 \leq 54n^3 \quad (n \geq 5) \quad (\mathbf{N \text{ is } 5})$$

- We want to make $4n^3 - 5n^2 + 50$ “look like” cn^3 .
- For the upper bound, can do anything that makes the function **larger**.
- For the lower bound, can do anything that makes the function **smaller**.

Exercise

True or False: $2n^2 + n \sin 3n = \Theta(n^2)$.

A: True

B: False

C: Impossible to determine

$$2n^2 + n \sin 3n \leq 2n^2 + n$$

Exercise

True or False: $2n^2 + n \sin 3n = \Theta(n^2)$.

A: True

B: False

C: Impossible to determine

$$2n^2 - n \leq 2n^2 + n \sin 3n \leq 2n^2 + n$$

Exercise

True or False: $2n^2 + n \sin 3n = \Theta(n^2)$.

A: True

B: False

C: Impossible to determine

Exercise

True or False: $f(n) = \Theta(n^2)$

Let

$$f(n) = \begin{cases} n^2 & \text{if } n \text{ is even} \\ 5 & \text{if } n \text{ is odd.} \end{cases}$$

A: True

B: False

**C: Impossible to
determine**

***Big-Oh (Big-O) and
Big-Omega***

Other Bounds

- $f = \Theta(g)$ means that f is both **upper** and **lower** bounded by factors of g .
- Sometimes we only have (or care about) upper bound or lower bound.
- We have notation for that, too.

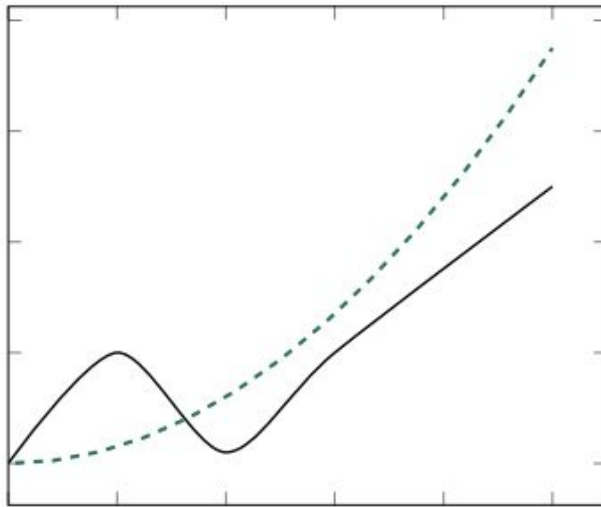
Big-O Notation, Informally

- Sometimes we only care about **upper bound**.
- $f(n) = O(g(n))$ if f “grows at most as fast” as g .
- **Examples:**
 - $4n^2 = O(n^{100})$
 - $4n^2 = O(n^3)$
 - $4n^2 = O(n^2)$ and $4n^2 = \Theta(n^2)$

Formal Definition

- We write $f(n) = O(g(n))$ if there are positive constants N and c such that for all $n \geq N$:

$$f(n) \leq c \cdot g(n)$$



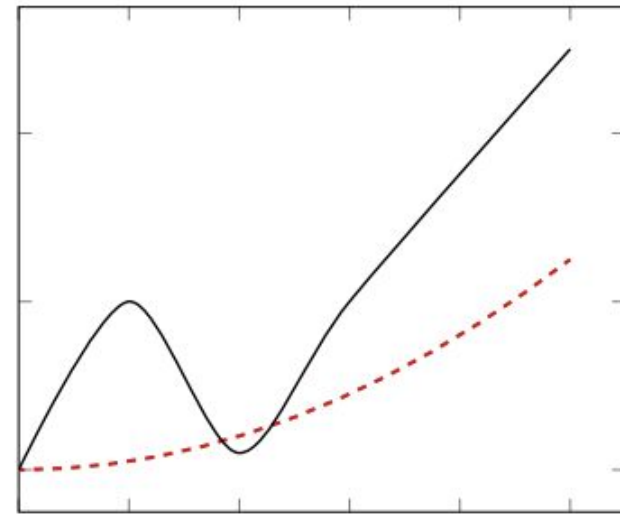
Big-Omega Notation, Informally

- Sometimes we only care about **lower bound**.
- $f(n) = \Omega(g(n))$ if f “grows at least as fast” as g .
- **Examples:**
 - $4n^{100} = \Omega(n^5)$
 - $4n^2 = \Omega(n)$
 - $4n^2 = \Omega(n^2)$ and $4n^2 = \Theta(n^2)$

Formal Definition

- We write $f(n) = \Omega(g(n))$ if there are positive constants N and c such that for all $n \geq N$:

$$c \cdot g(n) \leq f(n)$$



FUN FACT

“Omega” in Greek literally means: big O.
So translated, “Big-Omega” means “big big O”.

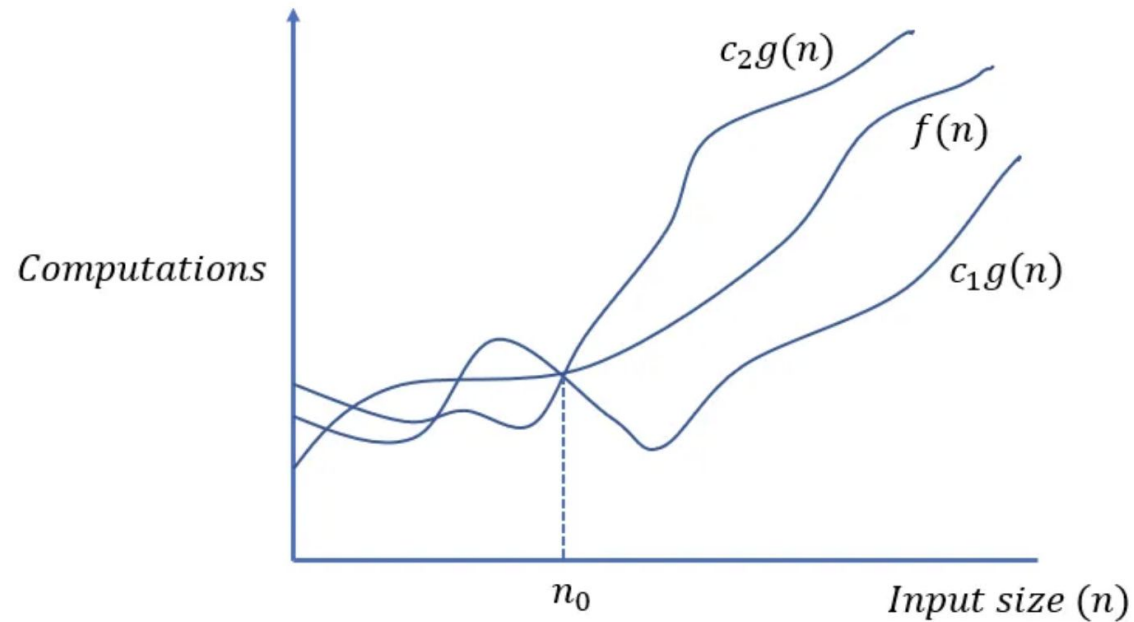
News

- Probably going to get a new room for the discussion.
 - I will update when/if it happens.

Last time: Formal definition of Θ

$$c_1g(n) \leq f(n) \leq c_2g(n)$$

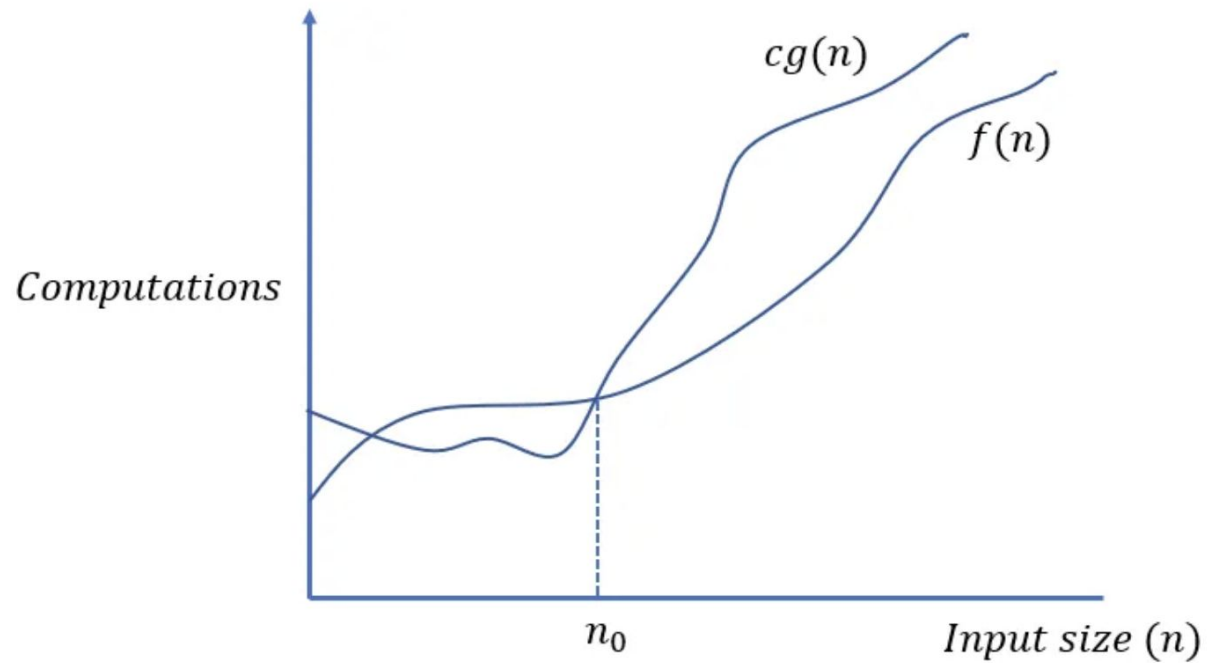
This can be visualised in the graph below.



Last time: Formal definition of O

$$0 \leq f(n) \leq cg(n) \quad \text{for } n \geq n_0$$

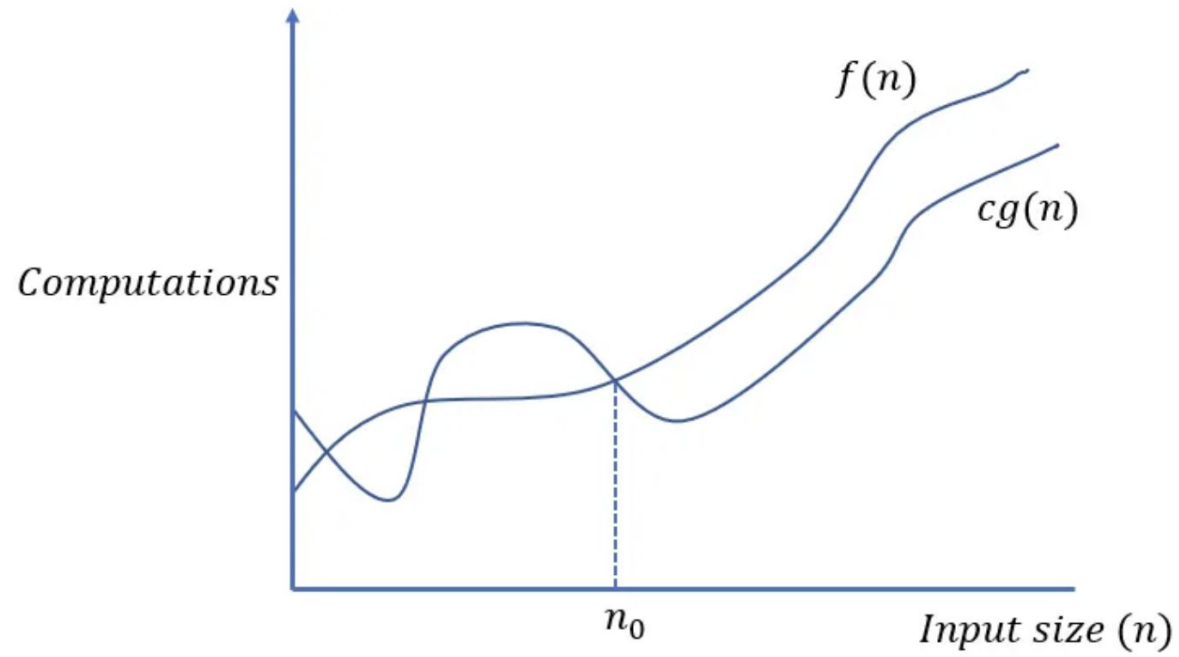
Equivalently, the graph of this upper bound is shown below.



Last time: Formal definition of Ω

$$0 \leq cg(n) \leq f(n) \quad \text{for } n \geq n_0$$

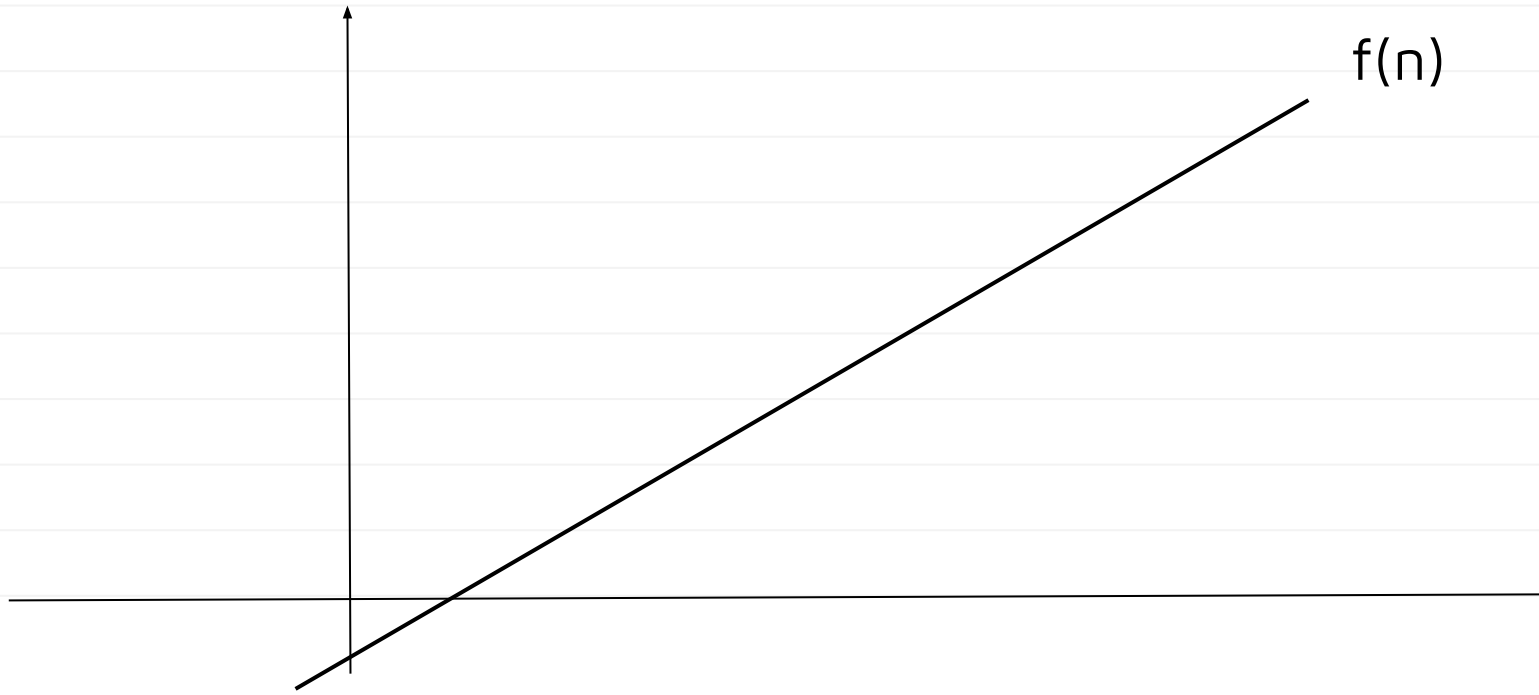
To gain a better understanding, let's look at this graph.



[Link](#)

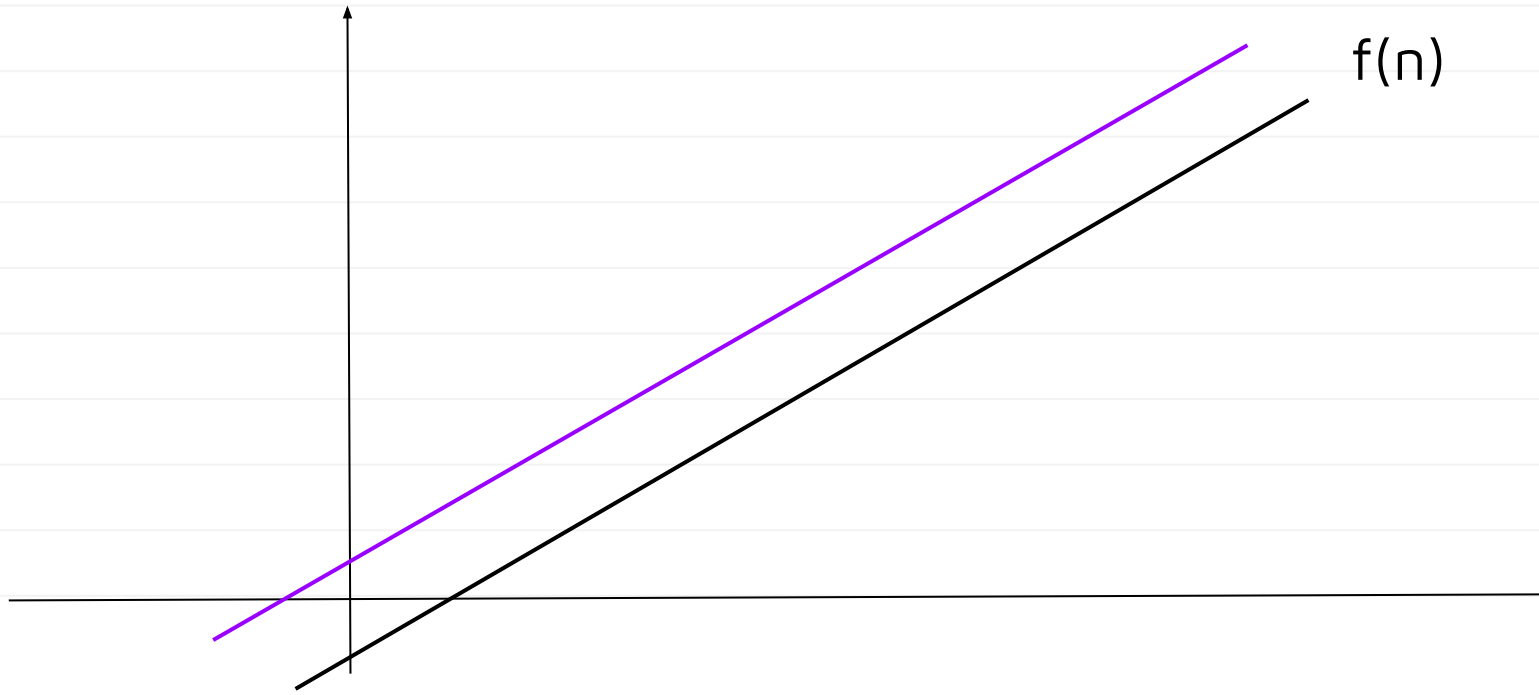
How many upper/lower bounds are there?

- If there is an upper/lower bound for a function, then there are infinitely many upper/lower bounds.



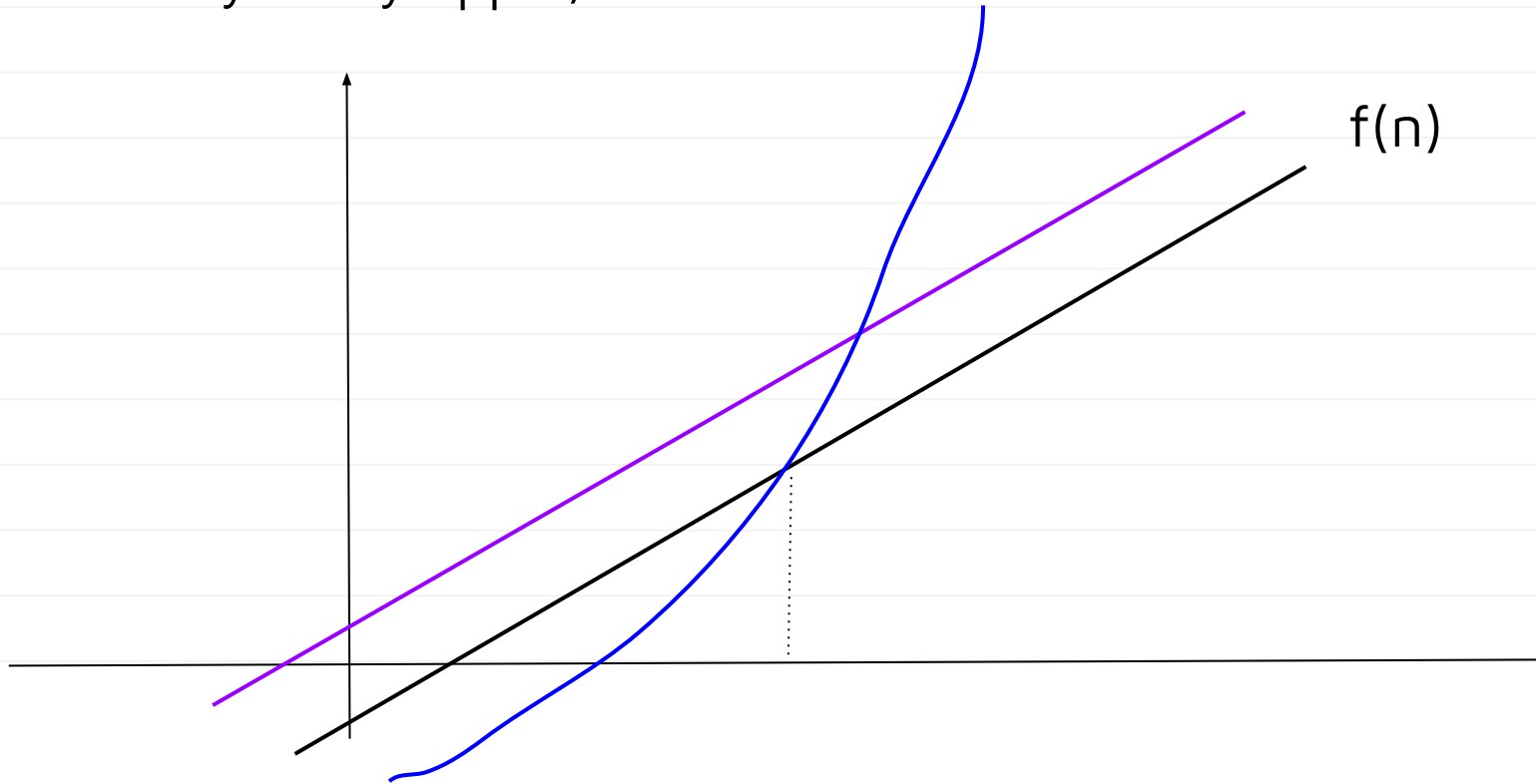
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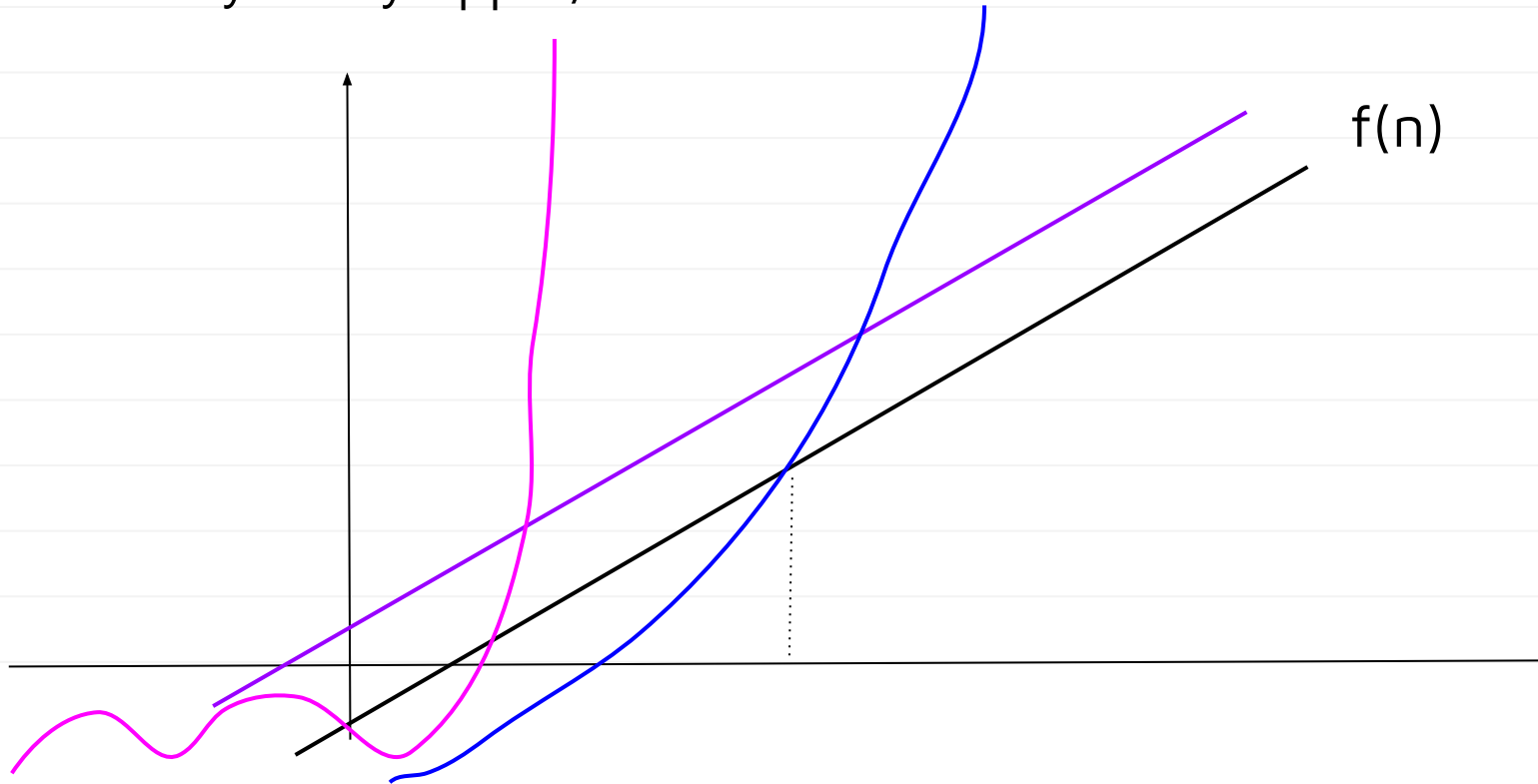
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Theta, Big-O, and Big-Omega

- If $f = \Theta(g)$ then $f = O(g)$ and $f = \Omega(g)$.
- If $f = O(g)$ and $f = \Omega(g)$ then $f = \Theta(g)$.
- Pictorially:
 - $\Theta \Rightarrow (O \text{ and } \Omega)$
 - $(O \text{ and } \Omega) \Rightarrow \Theta$

Analogies

- Θ is kind of like = $n^3 + 2$ *is kind of like* n^3 (not the exact =)
- O is kind of like $\leq$ (at most)
- Ω is kind of like $\geq$ (at least)

Practice: true/false

Let

$$f(n) = \begin{cases} n^2 & \text{if } n \text{ is even} \\ 5 & \text{if } n \text{ is odd.} \end{cases}$$

- True or False: $f(n) = O(n^2)$.
- True or False: $f(n) = \Omega(n^2)$.
- True or False: $f(n) = \Omega(1)$.

A: True, True, True

B: True, False, False

C: False, False, True

D: False, False, False

E: Something else

Practice: true/false

Let

$$f(n) = \begin{cases} n^2 & \text{if } n \text{ is even} \\ 5 & \text{if } n \text{ is odd.} \end{cases}$$

- True or False: $f(n) = O(n^2)$. (T)
- True or False: $f(n) = \Omega(n^2)$. (F)
- True or False: $f(n) = \Omega(1)$. (T)

Why Big-Oh

- Often used when another part of the code would dominate time complexity anyways.

Exercise: what is the complexity?

```
def foo(n):  
    for a in range(n**4):  
        print(a)  
  
    for i in range(n):  
        for j in range(i**2):  
            print(i + j)
```

Exercise: what is the complexity?

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$\Theta(n^4)$

```
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        for j in range(i**2):  
            print(i + j)
```

$\Theta(n^3)$ - ?

*maybe...I do not
care*

Exercise: what is the complexity?

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def foo(n):  
    for a in range(n**4):  
        print(a)
```

$\Theta(n^4)$

```
    for i in range(n):  
        for j in range(i**2):  
            print(i + j)
```

$O(n^3)$

enough

Big-Omega

- Often used when the time complexity will be **so large** that we don't care what it is, exactly.

Example: Big-Omega

```
best_separation = float('inf')
best_clustering = None

for clustering in all_clusterings(data):
    sep = calculate_separation(clustering)
    if sep < best_separation:
        best_separation = sep
        best_clustering = clustering

print(best_clustering)
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We have 2^n pairs

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We have 2^n pairs

$\Theta(?)$

$T(n) = \Omega(2^n)$

Other Notations

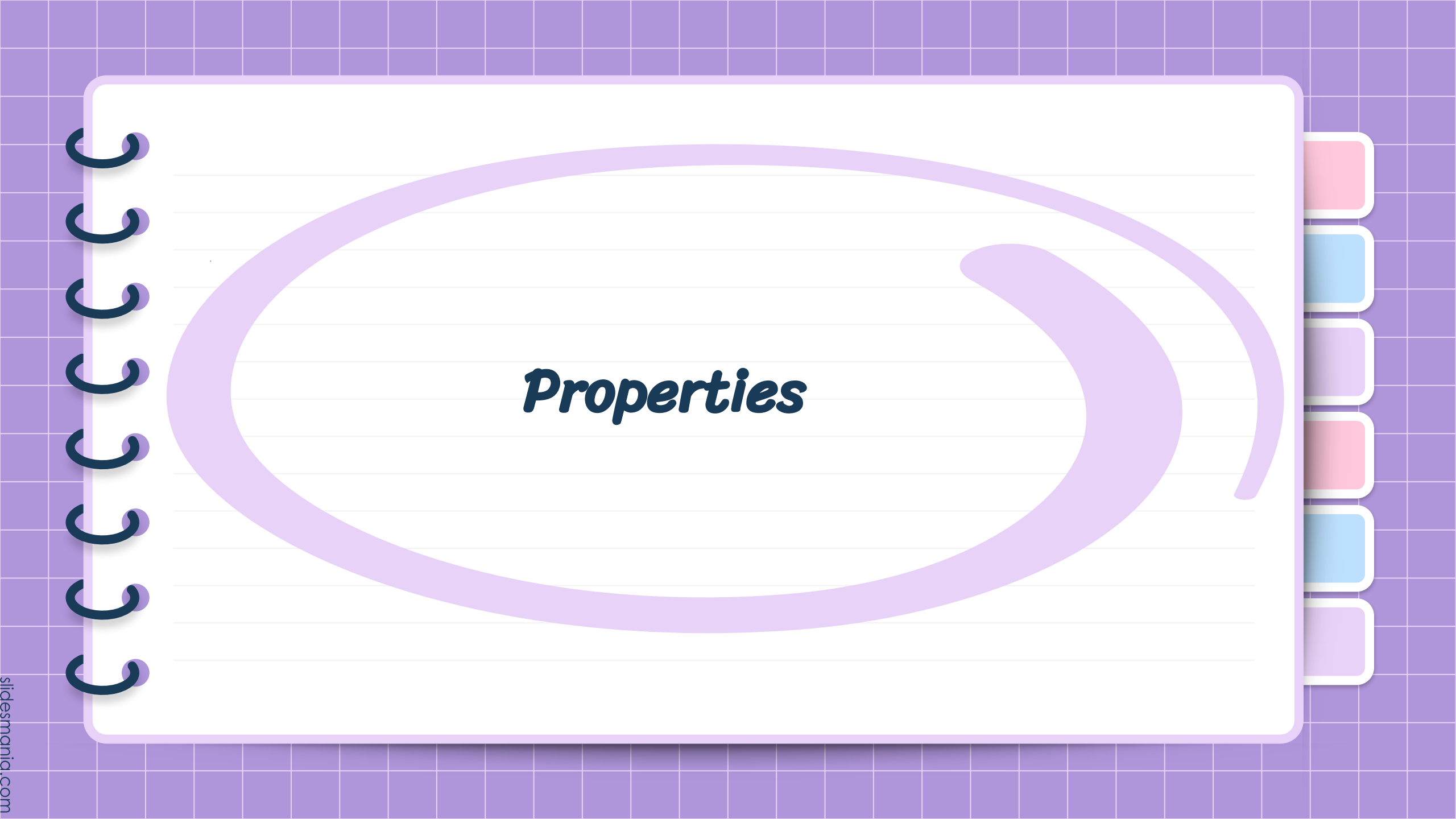
- $f(n) = o(g(n))$ if f grows “much slower” than g .
 - Whatever c you choose, eventually $f < cg(n)$.
 - Example: $n^2 = o(n^3)$

Other Notations

- $f(n) = o(g(n))$ if f grows “much slower” than g .
 - Whatever c you choose, eventually $f < cg(n)$.
 - Example: $n^2 = o(n^3)$
- $f(n) = \omega(g(n))$ if f grows “much faster” than g .
 - Whatever c you choose, eventually $f > cg(n)$.
 - Example: $n^3 = \omega(n^2)$

Other Notations

- $f(n) = o(g(n))$ if f grows “much slower” than g .
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 - Example: $n^2 = o(n^3)$
- $f(n) = \omega(g(n))$ if f grows “much faster” than g .
 - Whatever c you choose, eventually $f > cg(n)$.
 - Example: $n^3 = \omega(n^2)$
- **We won't really use these.**



Properties

Properties

- We don't usually go back to the definition when using Θ .
- Instead, we use a few basic ***properties***.

Properties of Θ

- **Symmetry:** If $f = \Theta(g)$, then $g = \Theta(f)$.
- **Transitivity:** If $f = \Theta(g)$ and $g = \Theta(h)$ then $f = \Theta(h)$.
- **Reflexivity:** $f = \Theta(f)$

Practice: T/F

- If $f = O(g)$ and $g = O(h)$, then $f = O(h)$.
- If $f = \Omega(h)$ and $g = \Omega(h)$, then $f = \Omega(g)$.
- If $f_1 = \Theta(g_1)$ and $f_2 = O(g_2)$, then $f_1 + f_2 = \Theta(g_1 + g_2)$.
- If $f_1 = \Theta(g_1)$ and $f_2 = \Theta(g_2)$, then $f_1 \times f_2 = \Theta(g_1 \times g_2)$.

A: F, F, F, T

B: F, T, F, F

C: T, F, F, T

D: T, T, T, T

E: None of the above

Practice: T/F

- If $f = O(g)$ and $g = O(h)$, then $f = O(h)$.
- If $f = \Omega(h)$ and $g = \Omega(h)$, then $f = \Omega(g)$.
- If $f_1 = \Theta(g_1)$ and $f_2 = O(g_2)$, then $f_1 + f_2 = \Theta(g_1 + g_2)$.
- If $f_1 = \Theta(g_1)$ and $f_2 = \Theta(g_2)$, then $f_1 \times f_2 = \Theta(g_1 \times g_2)$.

A: F, F, F, T

B: F, T, F, F

C: T, F, F, T

D: T, T, T, T

E: None of the above

Proving/Disproving Properties

- Start by trying to **disprove**.
- Easiest way: find a counterexample.
- **Example:** If $f = \Omega(h)$ and $g = \Omega(h)$, then $f = \Omega(g)$.
 - **False!** Let $f = n^3$, $h = n^2$ and $g = n^5$.

Proving the Property

- If you can't disprove, *maybe* it is true.
- **Example:**
 - Suppose $f_1 = O(g_1)$ and $f_2 = O(g_2)$.
 - Prove that $f_1 \times f_2 = O(g_1 \times g_2)$.

Step 1: State the assumption

- We know that $f_1 = O(g_1)$ and $f_2 = O(g_2)$.
- So there are constants c_1, c_2, N_1, N_2 so that for all $n \geq N_1$ and $n \geq N_2$:

$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

Use the assumption

Prove that $f_1 \times f_2 = O(g_1 \times g_2)$.

- Chain of inequalities, starting with $f_1 \times f_2$, ending with $\leq c g_1 \times g_2$.
- Using the following piece of information:

$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

Use the assumption

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$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

$$f_1(n) \times f_2(n) \leq \dots \leq c g_1 \times g_2$$

Use the assumption

Prove that $f_1 \times f_2 = O(g_1 \times g_2)$.

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- Using the following piece of information:

$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

$$f_1(n) \times f_2(n) \leq c_1 g_1(n) \times f_2(n) \quad (n \geq N_1)$$

Use the assumption

Prove that $f_1 \times f_2 = O(g_1 \times g_2)$.

- Chain of inequalities, starting with $f_1 \times f_2$, ending with $\leq c g_1 \times g_2$.
- Using the following piece of information:

$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

$$f_1(n) \times f_2(n) \leq c_1 g_1(n) \times f_2(n) \quad (n \geq N_1)$$

$$\leq c_1 g_1(n) \times c_2 g_2(n) \quad (n \geq N_1 \text{ and } n \geq N_2)$$

Use the assumption

Prove that $f_1 \times f_2 = O(g_1 \times g_2)$.

- Chain of inequalities, starting with $f_1 \times f_2$, ending with $\leq c g_1 \times g_2$.
- Using the following piece of information:

$$f_1(n) \leq c_1 g_1(n) \quad (n \geq N_1)$$

$$f_2(n) \leq c_2 g_2(n) \quad (n \geq N_2)$$

$$f_1(n) \times f_2(n) \leq c_1 g_1(n) \times f_2(n) \quad (n \geq N_1)$$

$$\leq c_1 g_1(n) \times c_2 g_2(n) \quad (n \geq N_1 \text{ and } n \geq N_2)$$

$$\leq c g_1(n) \times g_2(n) \quad (n \geq \max(N_1, N_2)) \text{ and } c = c_1 \cdot c_2$$

Analyzing Code

- The properties of Θ (and O and Ω) are useful when analyzing code.
- We can analyze pieces, put together the results.

Sums of Theta

- **Property:** If $f_1 = \Theta(g_1)$ and $f_2 = \Theta(g_2)$, then $f_1 + f_2 = \Theta(g_1 + g_2)$
- Used when analyzing sequential code.

Example

```
def foo(n):  
    bar(n)  
    baz(n)
```

- Say bar takes $\Theta(n^3)$,
- baz takes $\Theta(n^4)$.
- foo takes $\Theta(n^4 + n^3) = \Theta(n^4)$.
- baz is the **bottleneck**.

Products of Theta

- Property: If $f_1 = \Theta(g_1)$ and $f_2 = \Theta(g_2)$, then

$$f_1 \cdot f_2 = \Theta(g_1 \cdot g_2)$$

- Useful when analyzing **nested loops**.

Example

```
def foo(n):  
    for i in range(3*n + 4, 5n**2 - 2*n + 5):  
        for j in range(500*n, n**3):  
            print(i, j)
```


Example

```
def foo(n):  
    for i in range(3*n + 4, 5n**2 - 2*n + 5):  
        for j in range(500*n, n**3):  
            print(i, j)
```

A: $\Theta(n^5)$

B: $\Theta(n^2)$

C: $\Theta(n^4)$

D: $\Theta(n^3)$

E: Something else

Example

```
def foo(n):  
    for i in range(3*n + 4, 5n**2 - 2*n + 5):  
        for j in range(500*n, n**3):  
            print(i, j)
```

A: $\Theta(n^5)$

B: $\Theta(n^2)$

C: $\Theta(n^4)$

D: $\Theta(n^3)$

E: Something else

Careful!

If inner loop index **depends** on outer loop, you have to be more careful.

```
def foo(n):  
    for i in range(n):  
        for j in range(i):  
            print(i, j)
```



Asymptotic Notation Practicalities

In this part...

- Other ways asymptotic notation is used.
- Asymptotic notation slip ups.
- Downsides of asymptotic notation.

Not Just for Time Complexity!

- We most often see asymptotic notation used to express time complexity.
- But it can be used to express any type of growth!

Example: Combinatorics

- Recall: $\binom{n}{k}$ is number of ways of choosing k things from a set of n .
- How fast does this grow with n ? For fixed k :

$$\binom{n}{k} = \Theta(n^k)$$

- **Example:** the number of ways of choosing 3 things out of n is $\Theta(n^3)$.

Example: Central Limit Theorem

- Recall (DSC10): the CLT says that the sample mean has a normal distribution with standard deviation $\sigma_{\text{pop}}/\sqrt{n}$
- The **error** in the sample mean is: $O(1/\sqrt{n})$

Common Slip-ups

- Asymptotic notation can be used **improperly**.
 - Might be technically correct, but defeats the purpose.
- Don't do these in, e.g., interviews!

Slip-up #1

- Don't include constants, lower-order terms in the notation.
- **Bad**: $3n^2 + 2n + 5 = \Theta(3n^2)$.
- **Good**: $3n^2 + 2n + 5 = \Theta(n^2)$.
- It isn't *wrong* to do so, just defeats the purpose.

Slip-up #2

- Don't include base in logarithm.
- **Bad**: $\Theta(\log_2 n)$
- **Good**: $\Theta(\log n)$
- Why? $\log_2 n = c \cdot \log_3 n = c' \log_4 n = \dots$

Slip-up #3

- Don't misinterpret meaning of $\Theta(\cdot)$.
- $f(n) = \Theta(n^3)$ does **not** mean that there are constants so that $f(n) = c_3 n^3 + c_2 n^2 + c_1 n + c_0$.

Slip-up #4

- Time complexity is not a **complete** measure of efficiency.
- $\Theta(n)$ is not *always* “better” than $\Theta(n^2)$.
- Why?

Slip-up #4

- **Why?** Asymptotic notation “hides the constants”.
- $T_1(n) = 1,000,000n = \Theta(n)$
- $T_2(n) = 0.000001n^2 = \Theta(n^2)$
- But $T_1(n)$ is **worse** for all but really large n .

Main Idea

- Time Complexity is **not** the only way to measure efficiency, and it can be misleading.
- Sometimes even a $\Theta(2^n)$ algorithm is better than a $\Theta(n)$ algorithm, if the data size is small.

Caution

- To upper bound a fraction A/B , you must:
 - Upper bound the numerator, A .
 - Lower bound the denominator, B .
- And to lower bound a fraction A/B , you must:
 - Lower bound the numerator, A .
 - Upper bound the denominator, B .

Caution

- To upper bound a fraction A/B , you must:
 - Upper bound the numerator, A .
 - Lower bound the denominator, B .
- **Example:**
 - What is larger? $\frac{3}{5}$ or $\frac{4}{5}$?
 - $\frac{4}{5}$ since numerator is larger ($4 > 3$)
 - What is larger? $\frac{4}{3}$ or $\frac{4}{5}$?
 - $\frac{4}{3}$ since the denominator is smaller ($3 < 4$)

Caution

- To upper bound a fraction A/B , you must:
 - Upper bound the numerator, A .
 - Lower bound the denominator, B .
- To make a fraction **as large as possible**, you should:
 - Make the **numerator (A)** as **large as possible** → that's upper bounding A .
 - Make the **denominator (B)** as **small as possible** → that's lower bounding B .

Caution

- To make a fraction **as large as possible**, you should:
 - a. Make the **numerator (A)** as **large as possible** → that's upper bounding A.
 - b. Make the **denominator (B)** as **small as possible** → that's lower bounding B.

If **$A=3x+2$** and **$B=x+1$** and you know that **$1 \leq x \leq 4$**

What is the best upper bound for A/B ?

A: 6

B: $5/4$

C: 7

D: 14



Thank you!

Do you have any questions?

CampusWire!